Adaptive Dual-Domain Learning for Hyperspectral Anomaly Detection With State-Space Models

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***Abstract*— Recently, learning-based hyperspectral anomaly detection (HAD) methods have demonstrated outstanding performance, dominating mainstream research. However, the existing learning-based approaches still have two issues: 1) they rarely consider both the spatial sparsity and the interspectral similarity of hyperspectral imagery (HSI) simultaneously and 2) they treat all regions equally, often overlooking the importance of high-frequency information in HSI, which is key to distinguish background and anomalies. To address these challenges, we propose a novel HAD method based on spatial–spectral adaptive dual-domain learning, termed SSHAD. Specifically, we first introduce the spatial-wise selected state space module (SSSM) with linear complexity and the spectral-wise frequency division self-attention module (FDSM), which are combined in parallel to form a spatial–spectral block (SS-block). The SSSM captures the global receptive field by scanning the HSI spatial dimension through a multidirectional scanning mechanism. The FDSM extracts high-frequency and low-frequency information from the HSI via the discrete wavelet transform (DWT) and applies multiscale convolution and self-similarity attention respectively, ensuring the suppression of anomalies during the reconstruction process. This parallel structure enables the network to model cross-window connections, expanding its receptive field while maintaining linear complexity. We use the SS-block as the main component of our adaptive dual-domain learning network, forming SSHAD. Furthermore, we introduce a frequency-wise loss function to inhibit the reconstruction of high-frequency anomalies during background reconstruction. Comprehensive experiments conducted on four public datasets and two unmanned aerial vehicle (UAV)-borne datasets validate the superiority and effectiveness of SSHAD. The code will be publicly available at https://github.com/CZhu0066/SSHAD.**

***Index Terms*— Dual-domain learning, hyperspectral anomaly detection (HAD), Mamba, remote sensing, state space models (SSMs).**

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I. Introduction

HYPERSPECTRAL imagery (HSI) captures numerous contiguous spectral bands, offering detailed spectral information for each wavelength at every pixel [1]. Due to its extensive spectral range and high spectral resolution, HSI is beneficial for various applications, including agricultural assessment, mineral identification, urban planning, and environmental monitoring [2], [3], [4], [5], [6]. Notably, hyperspectral anomaly detection (HAD) is particularly crucial as it could identify anomalous objects that starkly contrast with the background environment [7], [8].

Generally, HAD methods assume that anomalous targets exhibit low probability occurrences and significant spectral differences from their surrounding backgrounds [9], [10]. Maximizing the utility of acquired HSI data and crafting an appropriate model are crucial steps in effective anomaly detection. However, the absence of prior information about anomalies poses challenges in achieving robust and adaptable detection [11]. Besides, the intricate and diverse background distribution in HSI complicates the distinguishing anomalies from the background [12].

A. Traditional Methods

The existing HAD methods could primarily be classified into three categories: statistical modeling-based methods, representation-based methods, and deep learning-based methods [13], [14]. Statistical models typically estimate the background from global or local pixels to detect anomalies. Reed–Xiaoli (RX) is a representative statistical anomaly detector that assumes the background conforms to a multivariate Gaussian distribution and identifies anomalies by computing the Mahalanobis distance [15]. This fundamental assumption supports various derivations of RX, such as the local RX (LRX) [16], the weighted RX [17], the kernel RX (KRX) [18], and the fractional Fourier entropy RX (FrFE-RX) [19]. These methods aim to overcome the limitations of traditional RX by modeling HSI background distributions more efficiently, which reduces sensitivity to anomalies. However, statistical modeling-based methods are limited by their reliance on assumptions about data distribution and background modeling.

To construct a more accurate background statistical distribution and enhance applicability in complex scenarios, researchers have proposed representation-based models, including collaborative representation (CR) [20], robust principal component analysis (RPCA) [21], and low-rank

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representation (LRR) [22], [23]. The CR detector (CRD) models background pixels as a linear combination of its neighboring pixels within adjacent concentric double windows, distinguishing anomalies as pixels that deviate significantly from this representation [24]. To mitigate the influence of anomalous pixels on background dictionaries construction, some improved CRDs have been proposed [25], [26], [27]. RPCA methods approximate the background as a low-dimensional subspace and separate outliers using sparse constraints, exemplified by methods such as RPCA-RX [28], low-rank and sparse matrix decomposition-based Mahalanobis distance (LSMAD) [29], and tensor principal component analysis (TPCA) [30]. LRR methods decompose HSI into a low-rank background component and a sparse anomaly component to detect anomalies [31], [32]. The graph and total variation regularized LRR (GTVLRR) reconstructs the background using multiple subspaces and utilizes residuals for anomaly detection, effectively handling complex intrinsic data [33]. Representation-based methods offer advantages in background modeling and anomaly separation, especially in scenarios with complex background distributions. However, the numerous linear combinations in CRDs and the time-consuming iterative optimization process in RPCAs contribute to high computational complexity, which limits their real-time performance. Additionally, LRRs often assume low-rank backgrounds or sparse anomalies, which may not hold in all practical applications.

B. Deep Learning Methods

Recently, with the advancement of deep learning, scholars have increasingly explored their applications in HAD, given the powerful feature extraction capabilities [34], [35], [36], [37]. Unlike traditional methods mentioned above, deep learning-based techniques eliminate the need for prior assumptions about the distribution of HSI and manual feature engineering. The prevailing deep learning-based methods could be broadly categorized into four groups: reconstruction-based methods, residual error-based methods, feature learning-based methods, and deep learning-supported methods [38].

Reconstruction-based methods detect anomalies by reconstructing the background, assuming that the reconstructed data follows a Gaussian distribution, while anomalies cannot be successfully reconstructed. Anomalies are then detected based on statistical differences, such as Mahalanobis distance. Common techniques include autoencoders (AEs), generative adversarial networks (GANs), and among other frameworks [39], [40]. Bati et al. [41] pioneer the application of AEs in HAD by reconstructing all samples in detection images. However, the limited effectiveness of background representation poses challenges for this method in complex scenes. Xiang et al. [42] propose a guidance module to improve background representation; however, the reliance on predefined guidance images limits the model's adaptability across varying scenes. To further enhance detection performance, some other methods have combined AEs and GANs to reconstruct the background component [43], [44]. Zhong et al. [45] propose BASGAN, a GAN-based method which combines class saliency map extraction algorithm with a background-anomaly separability-constrained loss function to enhance the separability between background and anomalies. However, BASGAN depends on the selection of pseudo-background and anomaly samples, which may impact separation performance if these pseudo-samples are inaccurately labeled or exhibit distribution inconsistencies. Xiao et al. [46] introduce S2DWMTrans, which utilizes a spatial–spectral dual-window mask transformer to enhance feature extraction and suppress anomaly reconstruction, thereby enhancing detection accuracy. Besides, DirectNet [47] introduces a sliding dual-window model and an adaptive-depth reconstruction network to primarily reconstruct background pixels, amplifying reconstruction errors for anomalous targets.

Residual-based methods detect anomalies by reconstructing background pixels and using the residual errors as anomaly scores, where larger residual errors indicate anomalies. Consequently, designing a network that effectively reconstructs the background while suppressing anomalies is crucial [48]. Wang et al. [49] propose Auto-AD, an autonomous HAD method that uses a fully convolutional AE with skip connections to reconstruct the background. Compared to existing AE-based methods, Auto-AD achieves anomaly detection automatically, without requiring manual parameter tuning or any preprocessing or postprocessing steps. Fan et al. [50] present a robust graph AE (RGAE) detector that addresses the limitations of traditional AE-based methods by incorporating $ł_{2,1}$-norm regularization and superpixel segmentation-based graph regularization.

Feature learning-based approaches involve utilizing neural networks to extract discriminative features directly from the original HSI data, typically followed by subsequent detector design for anomaly detection [51], [52], [53]. Li et al. [54] propose CNNAD, a supervised convolutional neural network designed for HAD, which utilizes convolutional layers to extract deep features and identify anomalies. However, supervised learning methods like CNNAD are limited by the availability of labeled data and training samples, which restricts their flexibility and generalizability in scenarios where spectral prior knowledge is absent. Lian et al. [55] utilize a gated transformer network (GT-HAD) to model the spatial–spectral similarity of HSI, separating background and anomaly features with an adaptive gating unit.

Deep learning-supported methods incorporate deep learning architecture as an intermediate tool, combining it with preprocessing step or traditional detectors for the final result [56]. Jiang et al. [57] present an unsupervised low-rank embedded network (LREN) designed specifically for HAD, effectively integrating feature extraction and anomaly estimation without the need for manual labeling. Wu and Wang [58] propose a transformer-based AE framework (TAEF) for nonlinear HAD, which utilizes both local and transitive global spatial information and demonstrates superior performance over current state-of-the-art methods.

C. Motivations and Contributions

The deep learning-based methods mentioned above still have the following issues. First, most existing methods focus

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on pixel-level detection [36], where limited receptive fields lead to insufficient learning of spatial distribution, making it difficult to fully integrate spectral and spatial information. Second, the aforementioned methods treat all regions equally and tend to overlook the differences between high-frequency anomalies and low-frequency backgrounds during the reconstruction process, resulting in suboptimal anomaly detection performance. Third, the computational complexity in global attention-based methods is quadratic to the spatial size, imposing high computational costs and posing challenges for extensive remote sensing applications. Existing strategies, such as reducing computational window sizes to enhance attention efficiency, sacrifice the receptive field size [59]. Additionally, most models exhibit instability across different datasets, and are prone to overfitting.

Naturally, one problem arises: How can we effectively combine spectral and spatial information to achieve efficient HAD while maintaining low computational complexity? Considering the recently proposed state space models (SSMs), i.e., Mamba [60], which effectively model long-range dependencies through state transformation and offer linear complexity and scalability in handling long sequence data. As a viable alternative to Transformers, Mamba [60] integrates a selection mechanism for adaptive input processing. Leveraging hardware-aware parallel scanning algorithms, Mamba demonstrates effectiveness and efficiency across various visual applications [61]. Inspired by this progress, several pioneering studies have introduced Mamba into remote sensing image processing [62], [63], [64]. These experimental results demonstrate that the Mamba architecture achieves performance comparable to advanced CNN and Transformer architectures in various remote sensing visual tasks, such as hyperspectral change detection [62], hyperspectral image classification [63], and infrared small target detection [64].

To this end, this article introduces an innovative adaptive dual-domain learning approach for self-supervised HAD, termed SSHAD. Specifically, we propose SSHAD based on residual-based methods, designing an HSI reconstruction network that reconstructs the pure background and detects anomalies through residual errors. First, we introduce the spatial-wise selected state space module (SSSM), which utilizes a novel multidirectional scanning mechanism to achieve a global receptive field with linear complexity, while capturing spatial sparsity. Next, we propose the spectral-wise frequency division self-attention module (FDSM) to capture interspectral similarity in HSI data. The FDSM extracts both high-frequency and low-frequency information via discrete wavelet transform (DWT), applying multiscale convolution and self-similarity attention, respectively, to suppress anomalies during the reconstruction process. To integrate spatial sparsity and interspectral similarity, we combine the SSSM and FDSM in parallel to form a spatial–spectral block (SS-block). This parallel structure allows the network to model cross-window connections and expand its receptive field without increasing computational complexity. The SS-block serves as the core component of a U-shaped encoder–decoder HSI reconstruction network. Additionally, we design a frequency-wise loss function to reduce the focus on high-frequency noise and anomalies, ensuring a cleaner background reconstruction.

The primary contributions of this study could be summarized as follows.

1) We propose a novel SSHAD for HAD, which effectively integrates spatial sparsity and interspectral similarity through a parallel design. To be specific, we detect anomalies via residual errors—the difference between original HSI and reconstructed HSI.
2) We integrate SSSM and FDSM in parallel to constitute an SS-block, which expands the network's receptive field while maintaining linear complexity and achieves the distinction between high-frequency anomalies from low-frequency background.
3) We introduce a frequency-wise loss function to suppress the reconstruction of high-frequency anomalies during background reconstruction, enhancing the network's ability to discriminate between the background and anomalies.
4) We conduct experiments on four public and two unmanned aerial vehicle (UAV)-borne HSI datasets, demonstrating the effectiveness and generalizability of SSHAD across various real-world scenarios, while achieving lower computational complexity and memory costs.

The rest of this study is summarized as follows. Section II elaborates on the preliminaries of Mamba and introduces the proposed SSHAD, detailing the specific structure of two well-designed modules and the loss function. Section III presents the experimental setups and corresponding comparison results. Finally, Section IV concludes this article and points out future research directions.

II. Proposed Method

In this section, we first introduce the preliminaries of SSMs and Mamba. Next, we provide an overview architecture of the proposed SSHAD method. Then, we offer an in-depth description of each module within the network. Finally, we discuss the frequency-wise loss function used during the training process.

A. Preliminaries

Classical SSM-based models, i.e., Structured State-Space Sequence Models (S4) [65], utilize linear time-invariant systems to map an 1-D input function or sequence $x(t) \in \mathbb{R}^L$ through an implicit latent state $h(t) \in \mathbb{C}^N$ to an output response $y(t) \in \mathbb{R}^L$. This process can be described using linear ordinary differential equations (ODEs), characterized by parameters such as the state transition matrix $\mathbf{A} \in \mathbb{C}^{N\times N}$, and projection parameters $\mathbf{B} \in \mathbb{C}^N$ and $\mathbf{C} \in \mathbb{C}^N$. The function could be expressed as follows:

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{1}$$

where $h'(t)$ represents the time derivative of $h(t)$.

To make the continuous-time systems in S4 more suitable for deep learning scenarios, the ODEs need to be converted to a discrete function. Specifically, they introduce a timescale

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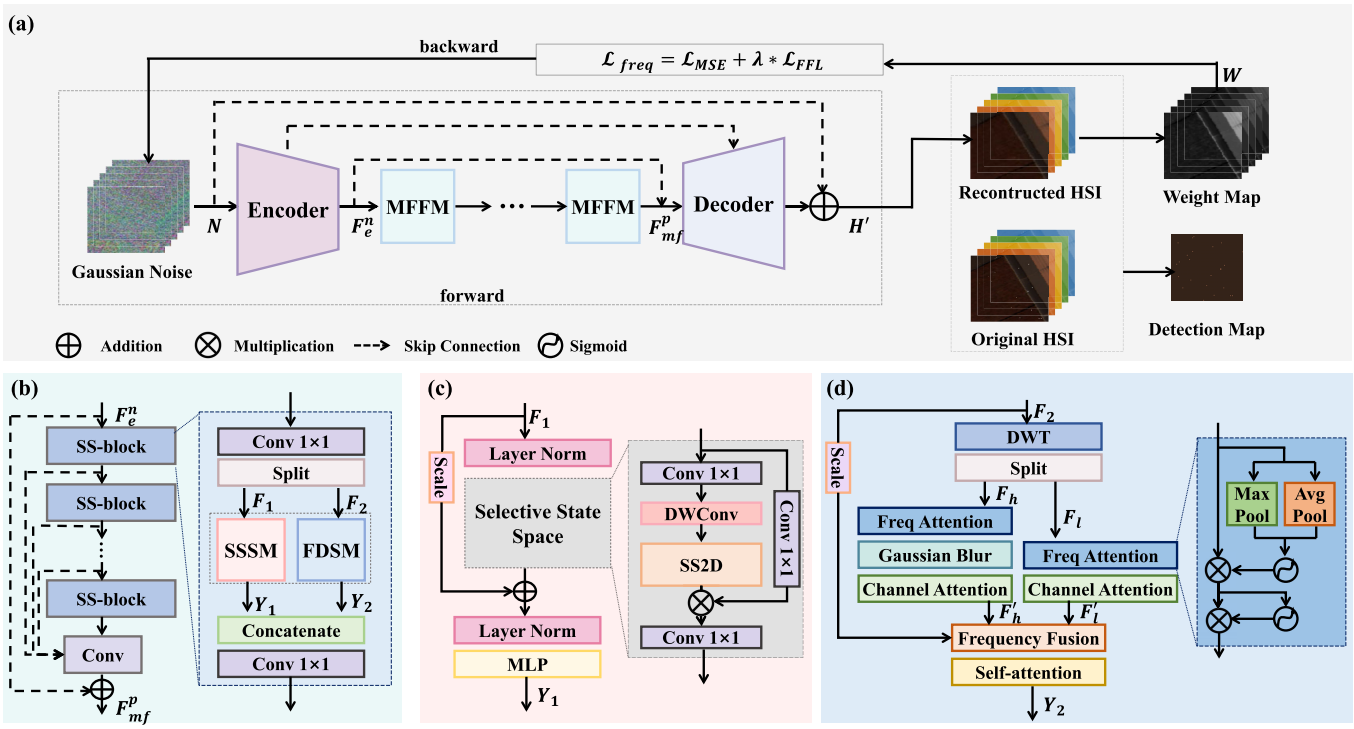


Fig. 1. Flowchart of the proposed SSHAD framework. (a) Overview of the proposed SSHAD framework. (b) Details of the MFFM and the SS-block. The spatial-wise SSSM and the spectral-wise FDSM are applied in parallel to form the SS-block. (c) Structure of the SSSM. (d) Structure of the FDSM.

parameter Δ to transform the continuous $\mathbf{A}$ and $\mathbf{B}$ into their discrete counterparts, denoted as $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$. This discretization formulation is achieved through zero-order hold (ZOH), which is described as follows:

$$\begin{aligned}\overline{\mathbf{A}} &= \exp(\Delta\mathbf{A}) \\ \overline{\mathbf{B}} &= (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B}.\end{aligned} \tag{2}$$

After discretization, (1) is reformulated as follows:

$$\begin{aligned}h_k &= \overline{\mathbf{A}}h_{k-1} + \overline{\mathbf{B}}x_k \\ y_k &= \mathbf{C}h_k\end{aligned} \tag{3}$$

where h_k represents the hidden state at the discrete time step k. Finally, after applying global convolution, the model's output can be calculated as follows:

$$\begin{aligned}y &= x * \overline{\mathbf{K}} \\ \overline{\mathbf{K}} &= \left(\mathbf{C}\overline{\mathbf{B}}, \mathbf{C}\overline{\mathbf{A}\mathbf{B}}, \ldots, \mathbf{C}\overline{\mathbf{A}}^{L-1}\overline{\mathbf{B}}\right)\end{aligned} \tag{4}$$

where $*$ represents the convolution operation, $\overline{\mathbf{K}}$ denotes a structured convolutional kernel, and L denotes the sequence length.

The Mamba [60] architecture enhances the traditional SSMs framework by introducing an input-dependent selective mechanism that dynamically filters relevant information from the input sequence. Unlike conventional SSMs, Mamba adapts the system parameters (Δ, $\overline{\mathbf{B}}$, $\mathbf{C}$) based on the evolving input data, allowing for more context-aware adjustments. Furthermore, hardware-aware optimization improves computational efficiency by optimizing GPU memory layout, accelerating both training and inference. By dynamically generating parameters in response to the input, Mamba effectively captures complex temporal relationships within sequences.

B. Overall Architecture

Fig. 1(a) presents an overview of the proposed SSHAD framework, which is based on a U-shaped encoder–decoder architecture. SSHAD detects anomalies by identifying residual errors during the background reconstruction process. This detection strategy is based on the following two hypotheses: 1) anomalies are typically sparse and small in size, making them difficult to reconstruct accurately and 2) it utilizes a HSI reconstruction network to reconstruct a pure background, where larger residual errors indicate anomalies and smaller errors are considered part of the background. Specifically, SSHAD can be divided into the following two stages.

1) Training Process: The original HSI, denoted as $\mathbf{H} \in \mathbb{R}^{H\times W\times B}$, is used for training, where H, W, and B represent the HSI's height, width, and number of bands, respectively. The background reconstruction network consists of three main components: an encoder, a multiscale feature fusion module (MFFM), and a decoder. The optimization of the network is achieved through a frequency-wise function that incorporates both adaptive mean squared error (mse) loss and focal frequency loss (FFL) [66].

a) Encoder: The network's input is an image filled with uniform Gaussian noise $\mathbf{N} \in \mathbb{R}^{H\times W\times B}$, which has the same dimensions as the original HSI $\mathbf{H}$. The SSHAD's encoder consists of n encoding layers, each including a convolutional layer, followed by a batch normalization (BN) layer and a

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rectified linear unit (ReLU) layer. The Gaussian noise $\mathbf{N}$ is passed through the encoder to generate the shallow feature map. The different shallow feature $\mathbf{F_e^i} \in \mathbb{R}^{(H/2^i)\times(W/2^i)\times 2^i C}$ could be described as follows:

$$\mathbf{F_e^i} = Y_E^{\mathbf{i}}(\mathbf{N}) \tag{5}$$

where $i \in \{1, 2, \ldots, n\}$ indicates the ith encoding layer, $Y_E^{\mathbf{i}}(*)$ denotes the function of the ith encoding layer, and the number of channels C is set to 32.

b) Multiscale feature fusion: Next, $\mathbf{F_e^n} \in \mathbb{R}^{(H/2^n)\times(W/2^n)\times 2^n C}$ is fed into p MFFMs, where each MFFM includes m SS-blocks and a convolutional layer. We utilize the SS-block for multiscale deep feature extraction and fusion to learn spatial sparsity and interspectral similarity simultaneously. Thus, the output multiscale feature $\mathbf{F_{ss}^j}$ of the jth SS-block is described as follows:

$$\begin{aligned}\mathbf{F_{ss}^1} &= Y_{\mathrm{SS}}^{\mathbf{1}}\left(\mathbf{F_e^n}\right)\\ \mathbf{F_{ss}^j} &= Y_{\mathrm{SS}}^{\mathbf{j}}\left(\mathbf{F_{ss}^{j-1}}\right)\end{aligned} \tag{6}$$

where $j \in \{2, 3, \ldots, m\}$ indicates the jth SS-block and $Y_{\mathrm{SS}}^{\mathbf{j}}(*)$ denotes the function of the jth SS-block. The multiscale feature fusion process in MFFM can be expressed as follows:

$$\mathbf{F_{ss}^m} = \sum_{j=1}^{m} w_j \cdot \mathrm{Conv}\left(\mathbf{F}_{\mathrm{ss}}^j\right) \tag{7}$$

where the weights w_j are learnable and dynamically adjusted during training to balance the contributions of each block. Together with the Conv operation, the multiscale features are adaptively aggregated across the different SS-blocks. Furthermore, the output of the MFFMs is formulated as follows:

$$\mathbf{F_{mf}^p} = Y_{\mathrm{MF}}^{\mathbf{k}}\left(\mathbf{F_{ss}^m}\right) \tag{8}$$

where $k \in \{1, 2, \ldots, p\}$ indicates the kth MFFM and $Y_{\mathrm{MF}}^{\mathbf{k}}(*)$ denotes the function of the kth MFFM. Notably, the number of encoder layers n, the number of MFFMs p, and the number of SS-blocks m are set as 3, 3, and 4, respectively. Additionally, further ablation experiments on the parameter settings will be presented in Section III-D.

c) Decoder: Afterward, the multiscale deep feature $\mathbf{F_{ss}^m}$ is fed into the decoder to reconstruct the background of the HSI, denoted as $\mathbf{H}' \in \mathbb{R}^{H\times W\times B}$. This decoder involves multiple decoding layers, each mirroring a corresponding component of the encoding layers. The output of the ith encoding layer is not only fed into the next encoding layer but is also directly connected to the corresponding decoding layers through skip connections. The process could be formulated as follows:

$$\mathbf{H}' = Y_D^{\mathbf{i}}\left(\mathbf{F_{mf}^p}, \mathbf{F_e^i}\right) \tag{9}$$

where $\mathbf{H}'$ represents the reconstructed HSI, $Y_D^{\mathbf{i}}(*)$ denotes the function of the ith decoding layer, and $\mathbf{F_e^i}$ represents the corresponding output from the encoding layer connected via skip connections. The output of the decoder integrates both multiscale and shallow features, enhancing reconstruction accuracy while improving training stability and convergence speed.

2) Detection Process: Finally, as the training process converges, we obtain the final reconstructed HSI $\tilde{\mathbf{H}}$, which represents the pure background component of the original HSI $\mathbf{H}$. Subsequently, we compare $\tilde{\mathbf{H}}$ with $\mathbf{H}$ to generate an anomaly detection map $\mathbf{D} \in \mathbb{R}^{H\times W}$ based on residual errors.

C. Spatial–Spectral Block

A well-designed network structure allows the network to model cross-window connections and expand its receptive field, which provides richer contextual information. Unlike obtaining feature maps solely through convolution and pooling operations [67], which can result in the loss of various levels of features during processing. Generally, shallower layers could acquire low-level features such as lines and textures, while deeper layers could acquire high-level features such as objects and shapes. Based on this, we combine the SSSM and FDSM in parallel to form the SS-block, as illustrated in Fig. 1(b).

To be specific, the shallow feature $\mathbf{F_e^n}$ is input into the SS-blocks. After a convolutional layer, $\mathbf{F_e^n}$ is split into two parts $\mathbf{F_1}$ and $\mathbf{F_2}$ along the channel dimension. The process is described as follows:

$$\mathbf{F_1}, \mathbf{F_2} = \mathrm{Split}\left(\mathrm{Conv}\left(\mathbf{F_e^n}\right)\right). \tag{10}$$

1) Selected State-Space Module: After the split operation, $\mathbf{F_1} \in \mathbb{R}^{(H/2^n)\times(W/2^n)\times 2^n C}$ is fed into the spatial-wise SSSM, as shown in Fig. 1(c). For the input $\mathbf{F_1}$, we apply layer normalization (LN) to stabilize the learning process and accelerate convergence by standardizing the input features across the batch dimension. This can be expressed as follows:

$$\mathbf{F_1}' = \mathrm{LN}(\mathbf{F_1}). \tag{11}$$

After the initial LN, $\mathbf{F_1}'$ is passed through a selective state space (SSS) component. The key component of the SSSM is SSS, which is designed based on the Select-Scan Structured State Space for Sequences (S6) block [60]. S6 serves as a fundamental component of the Mamba architecture, facilitating global receptive fields, dynamic weight allocation, and linear complexity. Its sequential scanning mechanism is primarily designed for natural language processing tasks. However, when applying S6 to visual tasks, challenges arise due to the noncausal and nonsequential nature of visual data. This results in significant image fragmentation and a loss of global receptive fields. To address this challenge, we introduce the 2D selective-scan (SS2D) in [61] to adapt to visual scenes and capture global spatial features in HSI, as shown in Fig. 2. Specifically, the 2-D image features are flattened into a 1-D sequence and scanned in four different directions to capture long-range dependencies using a discrete state-space equation. The resulting sequences are then merged through summation and reshaped to restore the 2-D structure. The SSS process is represented as follows:

$$\mathbf{F_s} = \mathrm{Conv}\left(\left(\mathrm{Conv}\left(\mathbf{F_1}'\right) + \mathrm{SS2D}\left(\mathrm{DWConv}\left(\mathrm{Conv}\left(\mathbf{F_1}'\right)\right)\right)\right.\right. \tag{12}$$

where $\mathbf{F_s}$ represents the output of SSS and DWConv refers to the depth-wise convolution. The selective mechanism in SS2D effectively suppresses anomaly signals, ensuring cleaner

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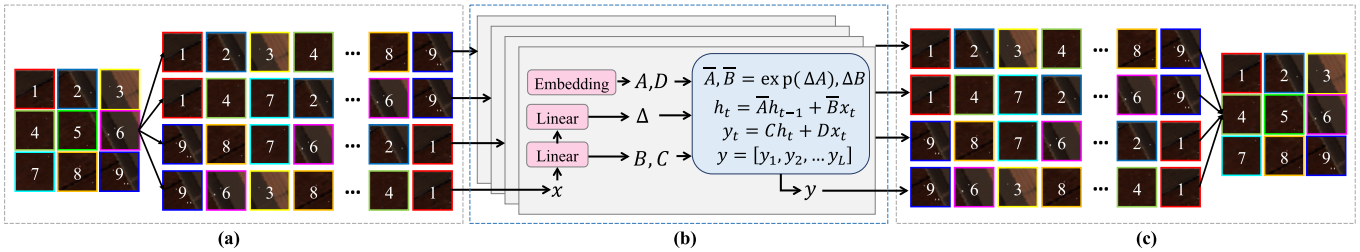


Fig. 2. Illustration of the SS2D on an HSI. (a) First, the input data is divided into multiple smaller patches. The patches are then scanned from four different directions to generate four feature sequences. (b) Each feature sequence is processed individually through the S6 block. (c) Finally, the four output sequences are merged to construct the final feature map.

background reconstruction. This mechanism captures spatial sparsity by highlighting distinct spatial patterns and outliers across different regions of the HSI, rather than weighing each region equally. By tuning the parameters of the S6 module, it retains globally consistent background features while filtering out spatially abrupt anomaly information. Moreover, depth-wise convolution is introduced to preserve essential spectral information while efficiently reducing noise, leading to a smoother background reconstruction process.

Notably, a scaling operation combines the input of SSSM $\mathbf{F_1}$ with the output of SSS $\mathbf{F_s}$ to retain the original input features. Subsequently, $\mathbf{F_s}$ undergoes LN once more. Afterward, the features pass through a multilayer perceptron (MLP) for additional nonlinear transformations and higher level feature extraction. The output of SSSM $\mathbf{Y_1}$ is formulated as follows:

$$\mathbf{Y_1} = \mathrm{MLP}(\mathrm{LN}(\mathrm{Scale}(\mathbf{F_1}) + \mathbf{F_s})). \tag{13}$$

2) Frequency Division Self-Attention Module: Similarly, $\mathbf{F_2} \in \mathbb{R}^{(H/2^n)\times(W/2^n)\times 2^n C}$ is fed into the frequency-wise FDSM, as shown in Fig. 1(d). To capture the differences in the HSI across different frequency domains, we introduce DWT [68] to decompose $\mathbf{F_2}$. The DWT splits the input $\mathbf{F_2}$ into four frequency sub-bands: the low-frequency band $\mathbf{F_l^0}$, and the high-frequency bands $\mathbf{F_{hh}}$, $\mathbf{F_{hv}}$, and $\mathbf{F_{hd}}$, representing horizontal, vertical, and diagonal details at a coarser resolution, respectively. The operation can be formulated as follows:

$$\left\{\mathbf{F_l^0}, \mathbf{F_{hh}}, \mathbf{F_{hv}}, \mathbf{F_{hd}}\right\} = \mathrm{DWT}(\mathbf{F_2}). \tag{14}$$

Next, we utilize a convolutional merging operation to combine the three high-frequency sub-bands into a unified component and use Pixel Shuffle to restore their size. The low-frequency component $\mathbf{F_l^0}$ is similarly processed through a convolution operation as follows:

$$\begin{aligned}\mathbf{F_h} &= \mathrm{Conv}(\mathrm{Concat}(\mathbf{F_{hh}}, \mathbf{F_{hv}}, \mathbf{F_{hd}}))\\ \mathbf{F_l} &= \mathrm{Conv}\left(\mathrm{Concat}\left(\mathbf{F_l^0}\right)\right).\end{aligned} \tag{15}$$

Furthermore, to suppress noise and enhance edge-related features, we introduce a Gaussian Blur to process the high-frequency component $\mathbf{F_h}$. We choose a light Gaussian blur kernel G with a size of 5×5, which is expressed mathematically as follows:

$$\mathbf{F_h^1} = G * \mathbf{F}_h. \tag{16}$$

Background features in HSI are typically smooth and continuous, corresponding to low-frequency components that capture structural information. In contrast, anomalies appear as small, high-frequency variations, containing detailed information that distinguishes them from the background. Considering these differences, we design a hybrid frequency attention module (FAM) that suppresses high-frequency components while emphasizing low-frequency components. This attention mechanism allows the model to selectively enhance background features and suppress anomalies, thereby improving the quality of the pure background reconstruction. The FAM first applies both adaptive max pooling and average pooling to the input. The outputs from the two pooling paths are then summed element-wise and passed through a sigmoid activation to generate the channel attention map. Finally, the input feature map is multiplied by this attention map to produce the output. The FAM function can be formulated as follows:

$$\mathrm{FAM} = \sigma(\mathrm{Conv}(\mathrm{MaxPool}(\mathbf{X}) + \mathrm{AvgPool}(\mathbf{X}))) \tag{17}$$

where σ means the sigmoid function, and MaxPool and AvgPool indicate the max pooling and average pooling, respectively. Based on this, when $\mathbf{F}_h^1$ and $\mathbf{F}_l$ pass through FAM, we can get the output as follows:

$$\mathbf{F}'_h, \mathbf{F}'_l = \mathrm{FAM}\left(\mathbf{F}_h^1\right), \mathrm{FAM}(\mathbf{F}_l). \tag{18}$$

$\mathbf{F}'_h$ and $\mathbf{F}'_l$ are then concatenated and fused via a weighted sum, which is added back to the original input feature map $\mathbf{F}_2$ through a scaling operation to retain the original details

$$\mathbf{F_f} = \alpha \mathbf{F}'_h + (1-\alpha)\mathbf{F}'_l + \mathrm{Scale}(\mathbf{F_2}) \tag{19}$$

where $\alpha \in [0, 1]$ is a learnable parameter controlling the contribution of high-frequency and low-frequency features. Subsequently, we apply self-attention (SA) to the residual output $\mathbf{F_f}$, refining spatial features and enhancing the model's ability to detect small anomalies in the index. The SA function can be formulated as follows:

$$\mathbf{Y_2} = \mathrm{SA}(\mathbf{F_f}). \tag{20}$$

Finally, $\mathbf{Y_1}$ and $\mathbf{Y_2}$ are combined and passed through a convolutional layer to obtain the multiscale deep feature $\mathbf{F_{ss}^m}$. This feature retains a residual connection with the initial input $\mathbf{F_e^n}$, which is represented as follows:

$$\mathbf{F_{ss}^m} = \mathrm{Conv}(\mathrm{Concat}(\mathbf{Y_1}, \mathbf{Y_2})) + \mathbf{F_e^n}. \tag{21}$$

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D. Loss Function

A suitable loss function accelerates network convergence and enhances the extraction of HSI's internal relationships. During the training process, we design a hybrid frequency-wise loss function by combining the adaptive mse loss [49] and FFL [66]. The implementation process is detailed as follows:

$$\mathbf{d}_{i,j} = \left\|\mathbf{h}_{i,j} - \mathbf{h}'_{i,j}\right\|_2 \tag{22}$$

where $\|\cdot\|_2$ represents the l_2-norm, $\mathbf{h}_{i,j}$ represents the pixel vector at position (i, j) in the original HSI $\mathbf{H}$, and $\mathbf{h}'_{i,j}$ denotes the pixel vector at the same position in the reconstructed HSI $\mathbf{H}'$. The reconstruction error at each position (i, j) is denoted by $\mathbf{d}_{i,j}$. Collectively, these errors form the reconstruction error map $\mathbf{E}$

$$\mathbf{E} = \left[\mathbf{d}_{1,1}, \ldots, \mathbf{d}_{1,W}; \cdots; \mathbf{d}_{H,1}, \ldots, \mathbf{d}_{H,W}\right]. \tag{23}$$

The adaptive weight map is calculated as follows:

$$\begin{aligned}\mathbf{w}_{i,j} &= \max(\mathbf{E}) - \mathbf{d}_{i,j} \\ \mathbf{W} &= \left[\mathbf{w}_{1,1}, \ldots, \mathbf{w}_{1,W}; \ldots; \mathbf{w}_{H,1}, \ldots, w_{H,W}\right].\end{aligned} \tag{24}$$

According to the weight map, we could define the adaptive mse loss as follows:

$$\mathcal{L}_{\text{mse}} = \text{mse}(\mathbf{WH}, \mathbf{WH}'). \tag{25}$$

The weight map allocates lower weights to areas with significant reconstruction errors, thereby minimizing the impact of anomalous pixels on the overall reconstruction errors. The pixel-wise calculation of mse loss often results in excessive smoothing and the loss of texture information during reconstruction due to the independent treatment of each pixel. To overcome this limitation, we introduce the FFL, which dynamically adjusts the spectrum weight matrix to adaptively focus on the frequencies that are difficult to reconstruct during training. This means that in each training iteration, the model can adjust its attention to different frequencies based on the current frequency loss, ensuring the reconstruction of the background while suppressing difficult-to-reconstruct high-frequency anomalies. The spectrum weight matrix $w_f(u, v)$, which provides the weight for each frequency component (u, v), is computed as follows:

$$w_f(u, v) = \left|F_r(u, v) - F_f(u, v)\right|^{\beta} \tag{26}$$

where $F_r(u, v)$ and $F_f(u, v)$ represent the real and generated image frequency components, respectively, and β is a scaling factor that controls the focus on high-frequency anomalies. The FFL is defined by applying the Hadamard product to the spectrum weight matrix and the frequency distance matrix as follows:

$$\mathcal{L}_{\text{FFL}} = \frac{1}{HW}\sum_{u=0}^{H-1}\sum_{v=0}^{W-1} w_f(u, v)\left|F_r(u, v) - F_f(u, v)\right|^2. \tag{27}$$

Next, we combine FFL with adaptive mse loss to form the hybrid frequency-wise loss function, integrating spatial and frequency information. The adaptive mse loss primarily focuses on pixel-level reconstruction, while FFL provides additional information in the frequency domain. The process is defined as follows:

$$\mathcal{L}_{\text{freq}} = \mathcal{L}_{\text{mse}} + \lambda\mathcal{L}_{\text{FFL}} \tag{28}$$

where the weight parameter λ regulates the trade-off between the two loss functions.

As the training process converges, we use the l_2-norm between the original HSI and the final reconstructed HSI to identify anomalies, i.e., residual errors. The detection map $\mathbf{D}$ is obtained as follows:

$$\mathbf{D} = \left\|\mathbf{h}_{i,j} - \tilde{\mathbf{h}}_{i,j}\right\|_2 \tag{29}$$

where $\mathbf{h}_{i,j}$ and $\tilde{\mathbf{h}}_{i,j}$ represent the pixels at location (i, j) of the original HSI $\mathbf{H}$ and the final reconstructed HSI $\tilde{\mathbf{H}}$, respectively. Therefore, the proposed SSHAD constructs an anomaly detector via residual errors. The entire process of the proposed SSHAD is described in Algorithm 1.

Algorithm 1 Overview of the Proposed SSHAD

Input: Original HSI $\mathbf{H} \in \mathbb{R}^{H\times W\times B}$, Gaussian noise $\mathbf{N} \in \mathbb{R}^{H\times W\times B}$, number of encoding layers n, number of SS-blocks m, number of decoding layers n
Output: Anomaly detection map $\mathbf{D} \in \mathbb{R}^{H\times W}$
Step 1: Training Process
while *training process* **do**
  1. Perform network forward process with equations (5) to (21);
  2. Update the weight map with equations (22) to (24);
  3. Compute the frequency-wise loss function with equations (25) to (28);
  4. Perform network backward process;
end
Step 2: Detection Process
Obtain the final reconstructed HSI $\tilde{\mathbf{H}} \in \mathbb{R}^{H\times W\times B}$;
Calculate the anomaly detection map $\mathbf{D}$ with equation (29);

III. Experiments and Discussions

In this section, we first introduce the datasets used in the experiments, including four public HSI datasets and two UAV-borne HSI datasets. Next, we provide a detailed explanation of the experimental settings. To evaluate the performance of the proposed SSHAD, we compare and analyze its detection results against nine state-of-the-art methods, using both qualitative and quantitative metrics. Furthermore, ablation experiments are conducted to validate the necessity of various components within the SSHAD framework. Finally, we discuss SSHAD's limitations and its applicable scenarios.

A. Data Description

The public datasets utilized in the experiments were captured by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensors. The datasets were publicly shared [69],

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TABLE I
INFORMATION OF THE FOUR PUBLIC HSI DATASETS

Datasets	Spatial Size	Bands	Spatial Resolution	Location	Sensor	Anomaly Type	Anomaly Pixels	Anomaly Proportion
HSI-I	100 × 100	205	7.1m	Los Angeles	AVIRIS	Flights	170	1.70%
HSI-II	100 × 100	204	17.2m	Texas Coast	AVIRIS	Buildings	67	0.67%
HSI-III	100 × 100	207	17.2m	Texas Coast	AVIRIS	Buildings	155	1.55%
HSI-IV	100 × 100	205	7.1m	Los Angeles	AVIRIS	Buildings	272	2.72%

including 13 HSI scenes. This article chooses four representative datasets which capture different scenes in the Texas Coast and Los Angeles, named HSI-I, HSI-II, HSI-III, and HSI-IV, respectively. The types of anomalies differ across datasets, with HSI-I capturing flights as anomalies, and the other three datasets detecting buildings of various sizes. The details of the four public datasets are presented in Table I and Fig. 3.

Due to the flexibility and low cost of UAVs, they are ideal for quickly locating and responding to postdisaster scenarios, such as monitoring urban waterlogging after heavy rains [70] or providing emergency response to forest fires [71]. In this study, we collected two UAV-borne HAD-Urban (UHAD-U) datasets, using a HyperspecI [72] visible and near-infrared sensor, mounted on a Chata5 UAV. The sensor is a snapshot spectral camera with a spectral range of 400–1000 nm, consisting of 61 channels at a 10 nm interval and an imaging frame rate of 47 frames/s. Data acquisition took place on July 10, 2023, over urban areas in Guangzhou, Guangdong Province, China, at a flight altitude of 100 m. The datasets cover two primary scenes: urban roads and an urban park. The details of the two UHAD-U datasets are presented in Table II. The first dataset, UHAD-U-I, was collected over a section of urban traffic road, consisting of 250 × 250 pixels and a spatial resolution of 0.053 m. The anomalous targets in this dataset primarily include two cars—one is relatively easy to identify, while the other is partially occluded by a tree, making it difficult to locate. The total number of anomalous pixels is 302, accounting for 0.48% of the entire image. The background of the road scene includes grass, trees, soil, and the urban traffic roads.

The second dataset, UHAD-U-II, represents a park scene where pedestrians standing on the grass are considered anomalous tiny targets. The size of UHAD-U-II is the same as UHAD-U-I. The total number of anomalous pixels is 34, accounting for 0.056% of the entire image. The background of the scene includes grass, trees, and the soil between them.

B. Experiment Setup

1) Comparative Methods: To validate the effectiveness of the SSHAD, comparative experiments are performed on nine representative algorithms which are frequently cited in the other related studies, including FrFE-RX [19], RPCA-RX [28], GTVLRR [33], TPCA [30], CNNAD [54], Auto-AD [49], LREN [57], DirectNet [47], and GT-HAD [55]. FrFE-RX is a statistical modeling-based method, while RPCA-RX, GTVLRR, and TPCA are representation-based. CNNAD, Auto-AD, LREN, DirectNet, and GT-HAD are deep learning-based methods. To ensure fairness in the experiments, the optimal parameter configurations for each method are used.



Fig. 3. False color and ground truth in four public HSI datasets used for experiments. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV.

2) Implementation Details: All deep learning-based experiments are conducted using PyCharm 2023.3.4 on a NVIDIA RTX 4090 GPU. The proposed SSHAD method is implemented using adaptive moment estimation for optimization, with a training process consisting of 100 epochs and a learning rate of $1e^{-4}$. The number of encoding and decoding layers n is set to 3, the number of MFFMs p is set to 3, and the number of SS-block m is set to 4. The parameter α in FDSM is set to 0.7 and the scaling factor β in FFL is set to 1. LREN is developed on the TensorFlow 1.14.0 platform, while other deep learning-based experiments are conducted on the PyTorch 2.1.1 platform. Additionally, nondeep learning comparative algorithms are executed in MATLAB R2020b on a computer equipped with 16 GB of RAM and an Intel Core i7-9700 CPU.

3) Evaluation Criteria: To fully validate the proposed SSHAD, we will employ both qualitative and quantitative evaluation criteria. The 2D ROC curves illustrate the relationship between detection probability P_D and false alarm rate P_F, effectively representing the model's detection accuracy. The 3D ROC curves add the threshold parameter τ, providing a more comprehensive analysis of performance across varying thresholds [73]. These curves can also generate two additional 2D ROC plots: P_D versus τ and P_F versus τ. The 2D ROC curve of (P_D, τ) evaluates the detection probability under varying thresholds, effectively assessing the model's anomaly detection performance. The 2D ROC curve of (P_F, τ) illustrates the false alarm rate over the same threshold range, providing insights into the model's capability to suppress background responses.

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TABLE II
INFORMATION OF THE TWO UHAD-U HSI DATASETS

Datasets	Spatial Size	Bands	Spatial Resolution	Location	Sensor	Anomaly Type	Anomaly Pixels	Anomaly Proportion
UHAD-U-I	250 × 250	61	0.053m	Guangzhou	HyperspecI	Cars	302	0.48%
UHAD-U-II	250 × 250	61	0.053m	Guangzhou	HyperspecI	Pedestrians	34	0.056%

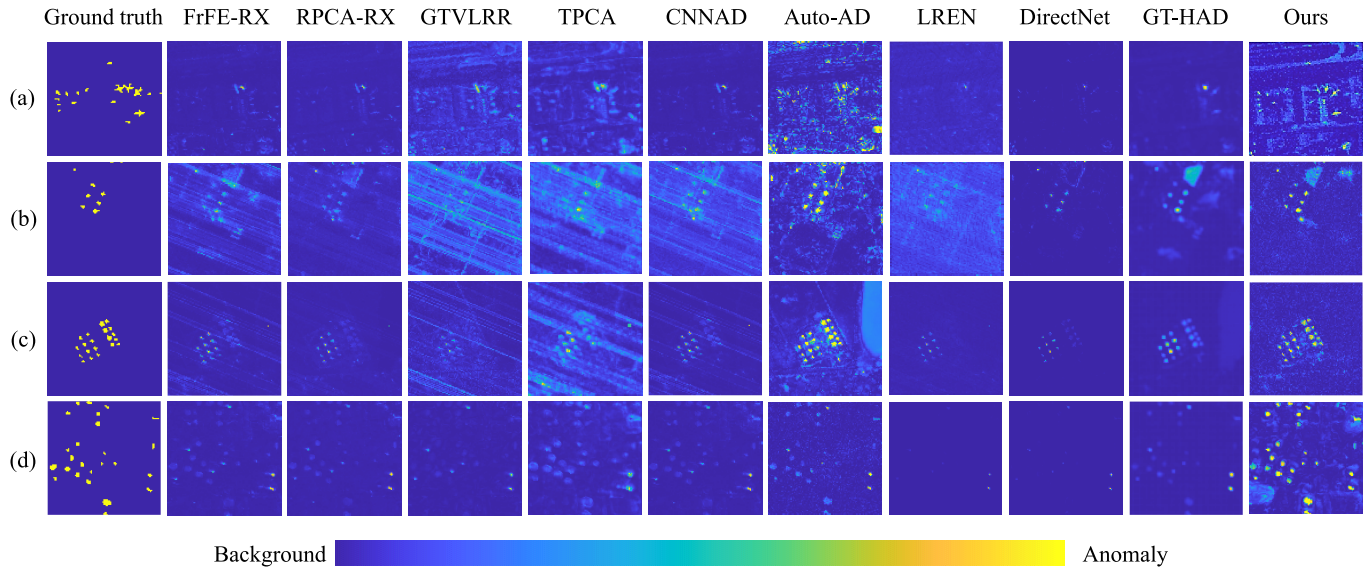


Fig. 4. Anomaly detection results across four public HSI datasets using various detectors. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV.

To comprehensively evaluate detection performance, we employ several AUC-based metrics beyond the conventional AUC(P_D, P_F), which represents the relationship between P_D and P_F. AUC(P_D, τ) and AUC(P_F, τ) are introduced to account for the influence of the threshold τ. Higher AUC(P_D, τ) indicates stronger detection performance across varying thresholds, while lower AUC(P_F, τ) reflects superior background suppression. Furthermore, the overall detection probability AUC_{OD} assesses algorithmic robustness, and the signal-to-noise probability ratio AUC_{SNPR} evaluates the detector's ability to distinguish anomalies from background noise. The equation is as follows:

$$\text{AUC}_{\text{OD}} = \text{AUC}(P_D, P_F) + \text{AUC}(P_D, \tau) - \text{AUC}(P_F, \tau) \tag{30}$$

$$\text{AUC}_{\text{SNPR}} = \frac{\text{AUC}_{(P_D,\tau)}}{\text{AUC}_{(P_F,\tau)}}. \tag{31}$$

Separability maps depict the data distribution, showing the separability of anomalies from the background. Each box represents the distribution of pixels within 10%–90%, with the horizontal line inside the box indicating the median pixel value. The vertical lines above and below each box show the pixel distribution between 90%–100% and 0%–10% ranges. The dashed lines at the top and bottom of each box indicate the maximum and minimum values of the data distribution. The distance between the red and green boxes represents the degree of separation between background and anomalies. Additionally, we evaluate the efficiency of different algorithms by considering model complexity and running time.

C. Detection Performance

1) Public Datasets: The comparative results across four public HSI datasets are visually depicted in Fig. 4. The anomalous targets in these datasets are all small-sized, typically consisting of 1 × 1 to 4 × 4 pixels. In particular, anomalies in the HSI-I and HSI-IV datasets exhibit a relatively scattered distribution, whereas those in the HSI-II and HSI-III datasets are more concentrated.

For example, in HSI-I as shown in Fig. 4(a), comparative methods struggle to locate anomalies. Although GTVLRR, TPCA, and Auto-AD could detect anomalous targets to some extent, they are limited to suppress background noise. In contrast, the proposed SSHAD effectively detects the anomalous target profiles and distributions. For the HSI-II dataset, shown in Fig. 4(b), GTVLRR and TPCA exhibit poor performance in detecting anomalies. Other methods, such as FrFE-RX, RPCA-RX, CNNAD, and LREN, can effectively identify the anomalous targets but are significantly influenced by the surrounding background, reducing the prominence of their detection effects. Although Auto-AD and GT-HAD highlight the targets, it also emphasizes some background components, leading to incomplete separation. DirectNet can detect targets while effectively suppressing background interference; however, its target detection range is limited, resulting in reduced precision. Obviously, SSHAD provides the most distinct detection maps compared to all

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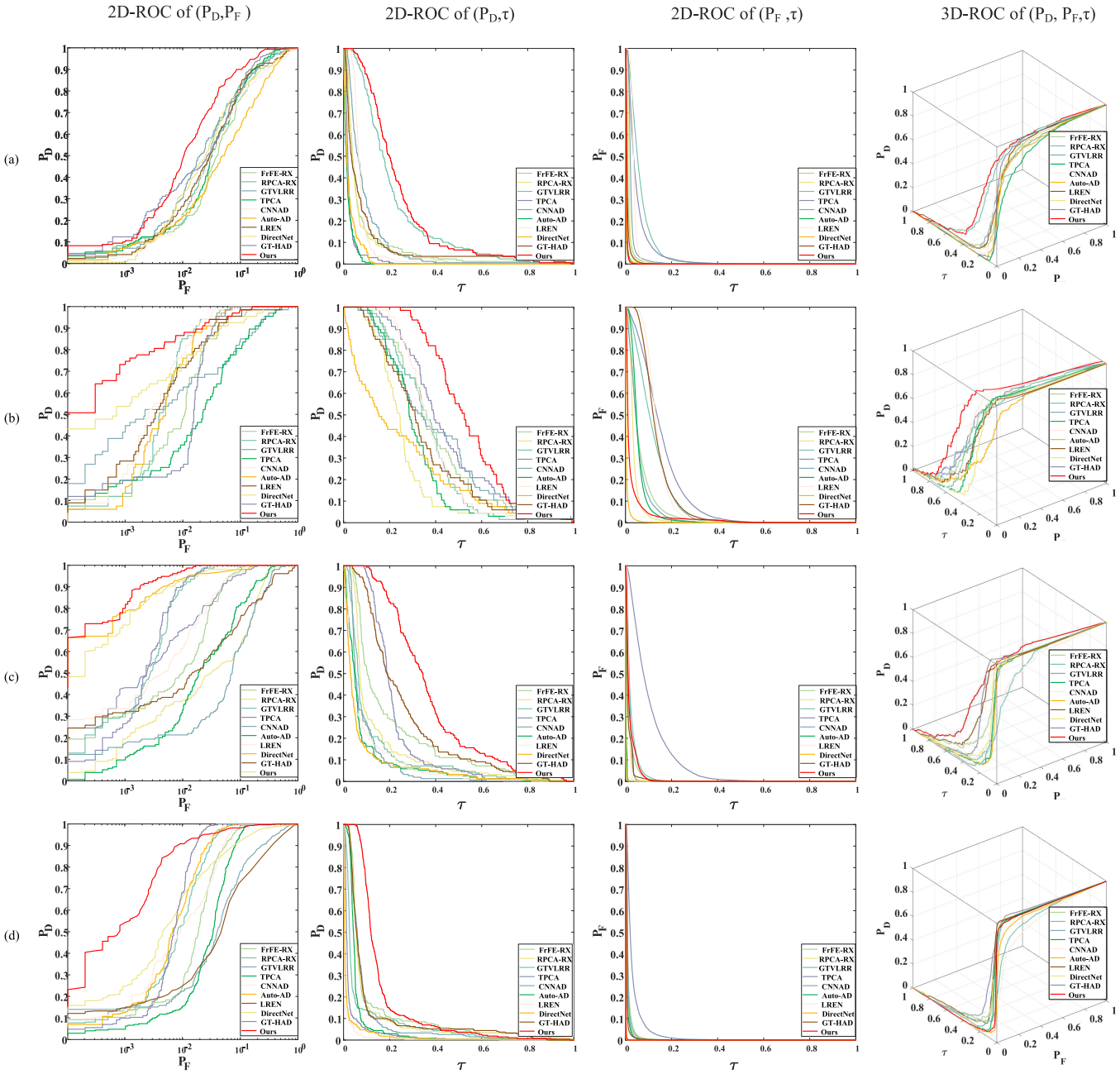


Fig. 5. ROC curves for diverse anomaly detectors across four public HSI datasets. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV.

other methods. Similarly, for the HSI-III dataset depicted in Fig. 4(c), detectors such as FrFE-RX, RPCA-RX, TPCA, CNNAD, Auto-AD, and GT-HAD exhibit closer alignment with the ground truth, whereas the remaining three methods show limited sensitivity to anomalous targets. However, Auto-AD struggles to accurately suppress the background while detecting anomalies, resulting in minimal distinction between the background and the anomalies. SSHAD outperforms the other methods, particularly performing significantly better than the deep learning-based methods LREN and DirectNet. The detection maps for the HSI-IV dataset from various detectors are shown in Fig. 4(d). Among these methods, GTVLRR, LREN, and DirectNet exhibit limited detection performance, failing to fully distinguish background from nuclear anomalies. In contrast, FrFE-RX, RPCA-RX, TPCA, CNNAD, Auto-AD, and GT-HAD achieve clearer and more satisfactory detection results. SSHAD, in particular, effectively detects anomalies while fully suppressing the background, enabling reliable extraction of anomalous target pixels.

As shown in Fig. 5 and Table III, 2D and 3D ROC curves, along with various AUC evaluation criteria, are used to quantitatively assess ten comparative methods across four public HSI datasets. In both the 2D ROC and 3D ROC curves, SSHAD consistently achieves higher P_D values than other methods, demonstrating superior performance. It effectively maintains high detection accuracy across a range of false alarm rates,

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TABLE III

AUC COMPARISON ACROSS FOUR PUBLIC HSI DATASETS USING DIFFERENT DETECTORS. BOLDFACE INDICATES THE BEST PERFORMANCE, AND UNDERLINING INDICATES THE SECOND-BEST PERFORMANCE

Datasets	AUC values	FrFE-RX	RPCA-RX	GTVLRR	TPCA	CNNAD	Auto-AD	LREN	DirectNet	GT-HAD	Ours
HSI-I	$\mathrm{AUC}(\mathrm{P_D},\mathrm{P_F})\uparrow$	0.9248	0.9276	0.9327	0.9297	0.9014	0.8742	0.9209	0.9191	0.9247	**0.9732**
	$\mathrm{AUC}(\mathrm{P_D},\tau)\uparrow$	0.0831	0.0747	0.2429	0.2211	0.0950	0.0276	0.0827	0.0316	0.0920	**0.2439**
	$\mathrm{AUC}(\mathrm{P_F},\tau)\downarrow$	0.0126	0.0167	0.0638	0.0504	0.0249	0.0029	0.0363	**0.0024**	0.0087	0.0047
	$\mathrm{AUC_{OD}}\uparrow$	0.9953	0.9856	1.1118	1.1004	0.9715	0.8989	0.9673	0.9483	1.0080	**1.2124**
	$\mathrm{AUC_{SNPR}}\uparrow$	6.5952	4.4731	3.8072	4.3869	3.8153	9.5172	2.2782	13.1667	10.5747	**51.8936**
HSI-II	$\mathrm{AUC}(\mathrm{P_D},\mathrm{P_F})\uparrow$	0.9815	0.9922	0.9448	0.9390	0.9824	0.9907	0.9805	0.9796	0.9833	**0.9930**
	$\mathrm{AUC}(\mathrm{P_D},\tau)\uparrow$	0.3943	0.2705	0.3649	0.4413	0.3905	0.3143	0.3850	0.2540	0.3482	**0.5381**
	$\mathrm{AUC}(\mathrm{P_F},\tau)\downarrow$	0.0630	0.0367	0.1163	0.1522	0.0558	0.0556	0.1371	**0.0178**	0.0266	0.0357
	$\mathrm{AUC_{OD}}\uparrow$	1.3128	1.2260	1.1934	1.2282	1.3171	1.2494	1.2284	1.2158	1.3050	**1.4954**
	$\mathrm{AUC_{SNPR}}\uparrow$	6.2587	7.3706	3.1376	2.8995	6.9982	5.6529	2.8082	14.2696	13.0902	**15.0728**
HSI-III	$\mathrm{AUC}(\mathrm{P_D},\mathrm{P_F})\uparrow$	0.9811	0.9957	0.8923	0.9409	0.9841	0.9951	0.9059	0.9964	0.9961	**0.9970**
	$\mathrm{AUC}(\mathrm{P_D},\tau)\uparrow$	0.2038	0.1446	0.1241	0.3776	0.0954	0.0900	0.0909	0.0823	0.2475	**0.2547**
	$\mathrm{AUC}(\mathrm{P_F},\tau)\downarrow$	0.0160	0.0138	0.0236	0.1032	0.0174	**0.0121**	0.0141	0.0202	0.0138	0.0192
	$\mathrm{AUC_{OD}}\uparrow$	1.1689	1.1265	0.9928	1.2153	1.0621	1.0730	0.9827	1.0585	1.2298	**1.2346**
	$\mathrm{AUC_{SNPR}}\uparrow$	12.7375	10.4783	5.2582	3.6589	5.4828	7.4380	6.4468	4.0743	**17.9348**	13.2656
HSI-IV	$\mathrm{AUC}(\mathrm{P_D},\mathrm{P_F})\uparrow$	0.9755	0.9871	0.9119	0.9671	0.9805	0.9844	0.8860	0.9729	0.9916	**0.9932**
	$\mathrm{AUC}(\mathrm{P_D},\tau)\uparrow$	0.1166	0.1034	0.0743	**0.1644**	0.0345	0.0611	0.0165	0.0241	0.1062	0.1282
	$\mathrm{AUC}(\mathrm{P_F},\tau)\downarrow$	0.0122	0.0115	0.0091	0.0322	0.0116	0.0058	0.0018	0.0008	0.0145	**0.0007**
	$\mathrm{AUC_{OD}}\uparrow$	1.0799	1.0790	0.9771	1.0993	1.0034	1.0397	0.9007	0.9962	1.0833	**1.1207**
	$\mathrm{AUC_{SNPR}}\uparrow$	9.5574	8.9913	8.1648	5.1056	2.9741	10.5345	9.1667	30.1250	7.3241	**183.1429**

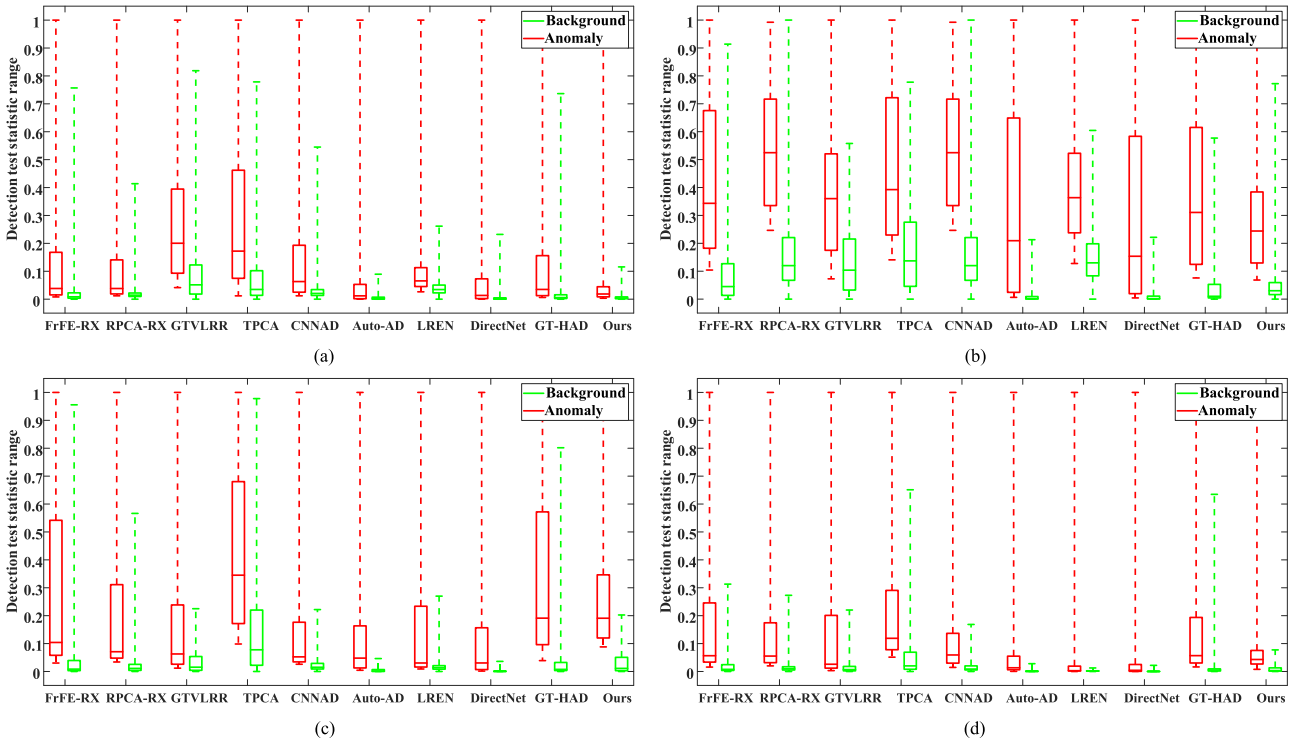


Fig. 6. Boxplots for diverse anomaly detectors across four public HSI datasets. (a) HSI-I. (b) HSI-II. (c) HSI-III. (d) HSI-IV.

proving its capability in suppressing background noise and accurately identifying anomalies. Results in Table III show that SSHAD achieves the highest AUC(P_D, P_F) values, indicating its exceptional ability to balance detection probability and false alarm rate. Specifically, compared to the well-known deep learning-based Auto-AD on the HSI-III dataset, SSHAD improves detection accuracy by 4.0%. This method also excels in AUC(P_D, τ), demonstrating robust detection performance across different thresholds. Specifically, SSHAD achieves higher values on the HSI-I dataset, with $\mathrm{AUC_{OD}}$ and $\mathrm{AUC_{SNPR}}$ values of 1.2124 and 51.8936, respectively, which show a clear advantage over other methods. In the

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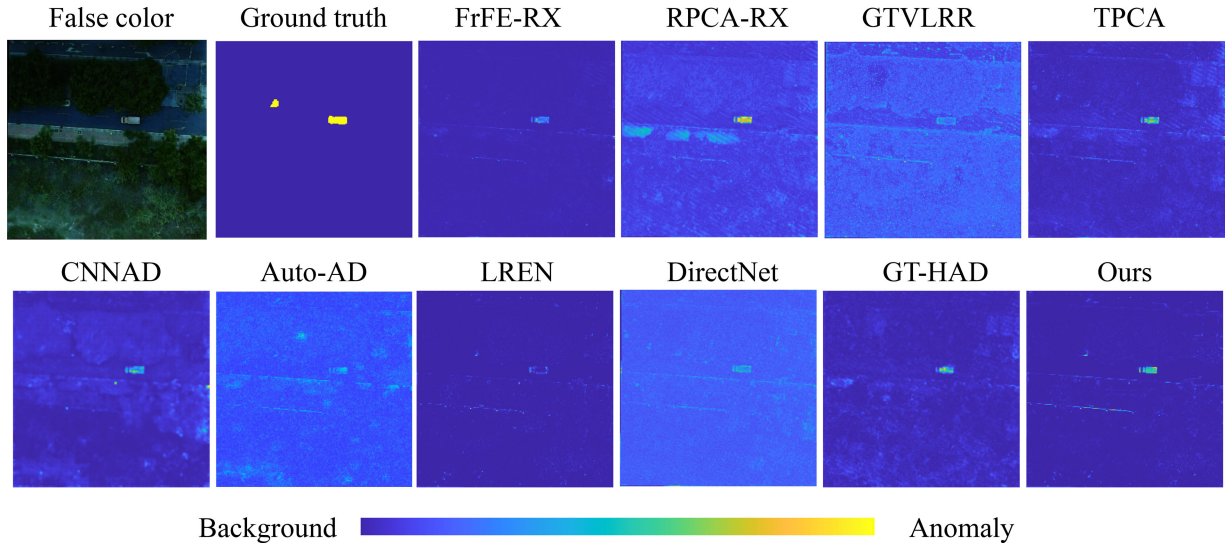


Fig. 7. Anomaly detection results across UHAD-U-I dataset using various detectors.

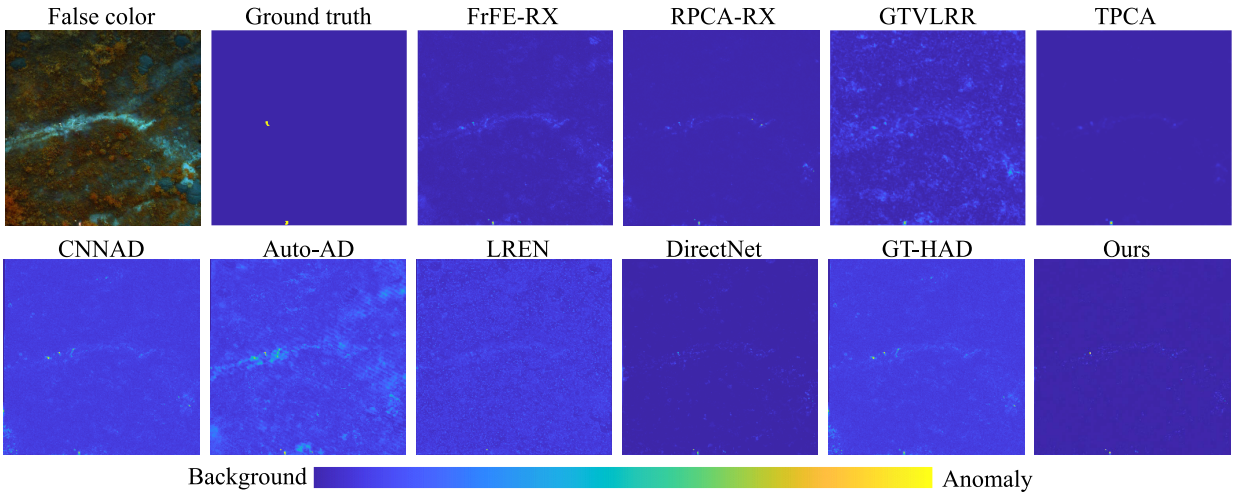


Fig. 8. Anomaly detection results across UHAD-U-II dataset using various detectors.

AUC_{SNPR} values for the HSI-III dataset, GT-HAD performs best, followed by SSHAD. It is evident that Auto-AD and DirectNet exhibit mixed results: while they successfully detect anomalous targets in the HSI-II, HSI-III, and HSI-IV datasets, their performance in HSI-I is notably poorer, leading to a lower average detection accuracy. Notably, among the deep learning-based methods, the proposed SSHAD outperforms CNNAD, Auto-AD, LREN, and DirectNet. This result highlights the effectiveness of incorporating multiscale spatial–frequency learning for enhancing background reconstruction and suppressing anomaly reconstruction. Additionally, SSHAD can achieve the lowest $AUC(P_F, \tau)$ in HSI-IV and the second-lowest in HSI-II, demonstrating a balanced trade-off between detection accuracy and false alarm rate. Overall, compared to mainstream methods, our algorithm shows acceptable performance across the four public HSI datasets.

Fig. 6 shows the background-anomaly separation performance for ten comparative methods, with red and green boxes representing the anomaly and background distributions, respectively. SSHAD achieves greater distance between these boxes, indicating superior background-anomaly separation. Additionally, the background distribution in SSHAD remains within a smaller range across all datasets, effectively suppressing background pixels. For example, in the HSI-II dataset, although Auto-AD and DirectNet show smaller background boxes, their separation performance is weaker. GTVLRR and TPCA exhibit larger anomaly boxes, causing overlap with the background and reducing detection accuracy. While other methods achieve reasonable separation, their background suppression could be improved. Overall, SSHAD demonstrates strong background-anomaly separation across four public HSI datasets, consistent with the earlier detection performance analysis.

2) UHAD-U Datasets: Figs. 7 and 8 present the visual evaluation results of ten comparative methods on the two UHAD-U datasets. For UHAD-U-I, there are two anomalous targets to be detected: one is a fully visible vehicle, which is relatively easy to detect, and the other is a partially obscured vehicle, with only the tail visible due to tree occlusion. This larger scale scene, with a more complex background

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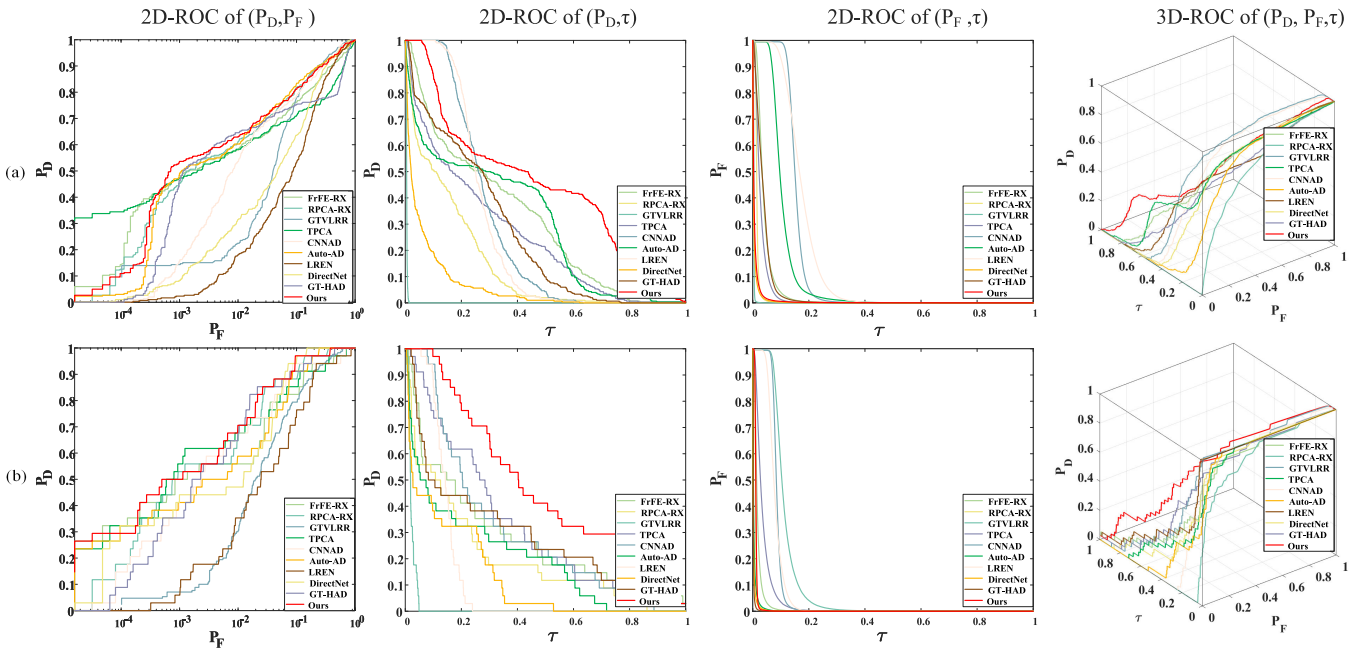


Fig. 9. ROC curves for diverse anomaly detectors across two UHAD-U HSI datasets. (a) UHAD-U-I. (b) UHAD-U-II.

TABLE IV

AUC COMPARISON ACROSS TWO UHAD-U HSI DATASETS USING DIFFERENT DETECTORS. BOLDFACE INDICATES THE BEST PERFORMANCE, AND UNDERLINING INDICATES THE SECOND-BEST PERFORMANCE

Datasets	AUC values	FrFE-RX	RPCA-RX	GTVLRR	TPCA	CNNAD	Auto-AD	LREN	DirectNet	GT-HAD	Ours
UHAD-U-I	$AUC(P_D, P_F) \uparrow$	0.8508	0.8950	0.8458	0.9231	0.8469	0.9195	0.7728	0.8461	0.8834	**0.9326**
	$AUC(P_D, \tau) \uparrow$	0.2604	0.2513	0.2406	0.1468	0.2574	0.2787	0.0052	0.2515	0.0609	**0.2864**
	$AUC(P_F, \tau) \downarrow$	0.0176	0.0407	0.1727	0.0154	0.0374	0.1539	**0.0038**	0.0285	0.0052	0.0124
	$AUC_{OD} \uparrow$	1.0936	1.1056	0.9137	1.0545	1.0669	1.0443	0.7742	1.0691	0.9391	**1.2066**
	$AUC_{SNPR} \uparrow$	14.7955	6.1744	1.3932	9.5325	6.8824	1.8109	1.3684	8.8246	11.7115	**23.0968**
UHAD-U-II	$AUC(P_D, P_F) \uparrow$	0.9413	0.9328	0.8931	0.9455	0.9488	0.9459	0.9080	0.9654	0.9549	**0.9753**
	$AUC(P_D, \tau) \uparrow$	0.2828	0.2086	0.0227	0.3287	0.3102	0.2072	0.1431	0.1351	0.2350	**0.4764**
	$AUC(P_F, \tau) \downarrow$	0.0176	0.0105	0.0114	0.0355	0.0865	0.0057	0.0798	0.0050	**0.0045**	0.0078
	$AUC_{OD} \uparrow$	1.2065	1.1309	0.9044	1.2387	1.1725	1.1474	0.9712	1.0955	1.2454	**1.4439**
	$AUC_{SNPR} \uparrow$	16.0682	19.8667	1.9912	9.2592	3.5861	36.3509	1.7932	27.0200	52.2222	**61.0769**

distribution and partially obscured anomalies, poses a challenge to the detection accuracy of the algorithms. Clearly, both LREN and GTVLRR perform poorly in detecting these two targets. FrFE-RX, RPCA-RX, TPCA, CNNAD, DirectNet, and GT-HAD successfully detect the unobscured vehicle but are unable to effectively detect the obscured target. While Auto-AD roughly identifies the regions where both anomalous targets are located, its detection accuracy is not high, and background suppression requires improvement. RPCA-RX detects the fully visible vehicle most clearly, though part of the road background is mistakenly identified as an anomaly, indicating insufficient suppression. Overall, the proposed SSHAD performs best, demonstrating excellent resistance to tree occlusion, road, and lawn interference, while also showing strong capability in vehicle detection.

For UHAD-U-II in Fig. 8, the background in this scene consists mainly of grass, trees, and a walkway, with the anomalous targets being two people: one standing on the walkway in the middle of the scene, and the other standing on the grass at the bottom of the scene. The detection challenge in this large-scale scene arises from the small size of the anomalous targets, which consist of only 34 pixels out of the entire HSI's spatial size of 250 × 250, representing only 0.056% of the total. Accurately detecting such small anomalies poses a challenge for algorithm performance. Clearly, LREN provides only a faint indication of the anomalies. TPCA can only detect the small target at the bottom of the scene and completely overlooks the small target in the middle. GTVLRR and Auto-AD exhibit high levels of background interference while detecting the targets. In contrast, FrFE-RX, RPCA-RX, CNNAD, DirectNet, and GT-HAD offer better background noise suppression, although some parts of the anomalies remain underdetected or indistinct. The proposed method achieves clear detection of the anomalies with superior background suppression, closely consistent with the ground truth.

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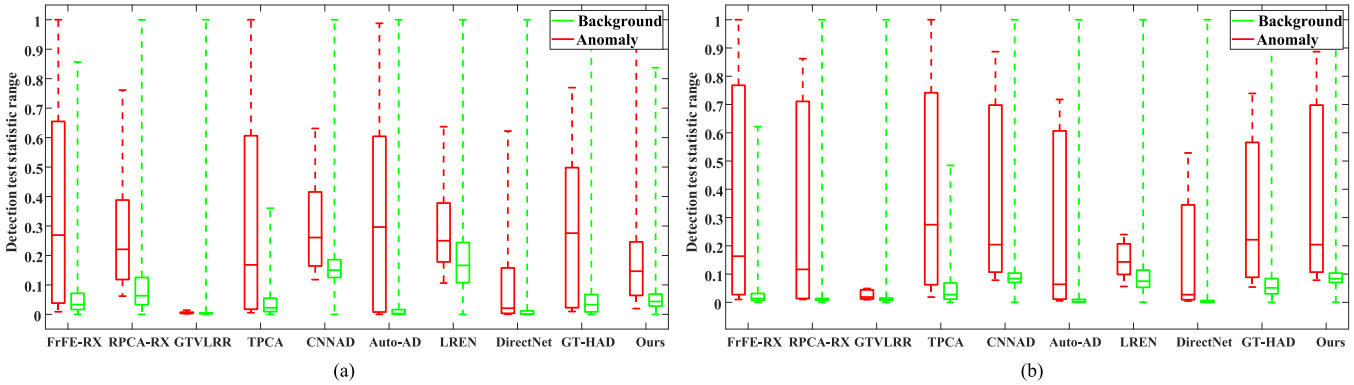


Fig. 10. Boxplots for diverse anomaly detectors across two UHAD-U HSI datasets. (a) UHAD-U-I. (b) UHAD-U-II.

TABLE V
ABLATION EXPERIMENTS OF SSHAD ON SIX DATASETS

AE	SSSM	FDSM	HSI-I	HSI-II	HSI-III	HSI-IV	UHAD-U-I	UHAD-U-II
✓			0.9245	0.9102	0.9664	0.9760	0.8954	0.9326
✓	✓		0.9621	0.9741	0.9966	0.9795	0.9263	0.9551
✓		✓	0.9521	0.9835	0.9962	0.9876	0.9142	0.9612
✓	✓	✓	**0.9732**	**0.9930**	**0.9970**	**0.9932**	**0.9326**	**0.9753**

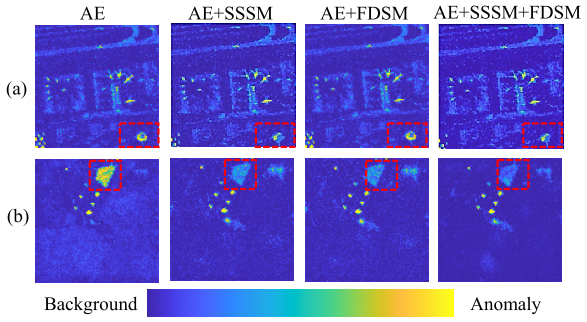


Fig. 11. Visual ablation experiments of the SSHAD method on two public HSI datasets. (a) HSI-I. (b) HSI-II. Red boxes mark parts of the background that are prone to misidentification as anomalies.

TABLE VI
COMPARISON OF PARAMETER p AND m ON HSI-III

Number of MFFMs p	Number of SS-blocks m	AUC (P_D, P_F)	Params/M	FLOPs/G
3	3	**0.9970**	**0.224**	**0.218**
	4	0.9965	0.289	0.259
	6	**0.9970**	0.419	0.340
4	3	0.9963	0.293	0.267
	4	0.9965	0.379	0.321
	6	0.9947	0.552	0.430
6	3	0.9958	0.434	0.367
	4	0.9961	0.564	0.449
	6	0.9960	0.823	0.611

Fig. 9(a) presents the 2D ROC and 3D ROC curves for the UHAD-U-I dataset. The red curve, representing the proposed method, shows moderate performance at low P_F values, but once P_F exceeds 0.001, its detection performance improves significantly, outperforming other comparison methods. In contrast, LREN, GTVLRR, DirectNet, and CNNAD perform relatively poorly, which is consistent with the detection maps in Fig. 7 and the AUC values in Table IV. In terms of AUC values, the proposed method achieves the highest AUC(P_D, P_F), AUC(P_D, τ), AUC_{OD}, and AUC_{SNPR}, indicating its excellent detection performance. Fig. 9(b) presents the 2D ROC and 3D ROC curves for the UHAD-U-II dataset. The red line representing the proposed SSHAD consistently demonstrates outstanding detection performance. Table IV provides a comparative analysis of the AUC performances of various detectors. Overall, SSHAD excels in most evaluation metrics. In the AUC(P_D, P_F) metric, SSHAD achieves the highest value, surpassing other methods, with DirectNet following closely at 0.9654. In the AUC(P_F, τ) metric, GT-HAD leads with the lowest value of 0.0045, indicating effective suppression of false alarms. SSHAD closely follows with a value of 0.0078, demonstrating a low false alarm rate. For the AUC_{OD} metric, SSHAD again achieves the highest value of 1.4439, showing superior overall detection performance. Additionally, in terms of AUC_{SNPR}, SSHAD achieves a significant advantage with a value of 61.0769, highlighting its strong signal-to-noise ratio recovery capability.

In the separability maps shown in Fig. 10(a), the background and anomaly boxes for FrFE-RX, GTVLRR, TPCA, Auto-AD, LREN, DirectNet, and GT-HAD do not exhibit significant separation. Although RPCA-RX and CNNAD achieve some separation between the background and anomaly boxes, the width of the background box remains large, indicating insufficient background suppression. In contrast, the proposed method achieves the greatest separation between the background and anomaly boxes which have the shortest background box width. In the separability maps shown in

TABLE VII
COMPARISON OF DIFFERENT CONVOLUTIONS ON HSI-III

Kernel size	Padding	AUC (P_D, P_F)	Params/M	FLOPs/G
1×1	0	0.9970	0.224	0.218
3×3	1	0.9958	0.814	0.586
5×5	2	0.9908	1.993	1.324
7×7	3	0.9901	3.763	2.430
9×9	4	0.9893	6.122	3.904

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TABLE VIII
AUC(P_D, P_F) COMPARISON OF ADDITIONAL TWO PUBLIC HSI DATASETS

Datasets	FrFE-RX	RPCA-RX	GTVLRR	TPCA	CNNAD	Auto-AD	LREN	DirectNet	GT-HAD	Ours
HYDICE	0.9906	0.9951	0.9943	0.8135	0.9826	0.9959	0.9636	0.9924	0.9932	**0.9960**
PAVIA	0.8441	0.9599	0.9706	0.9583	0.8804	0.9599	0.9401	0.9448	0.9538	**0.9813**

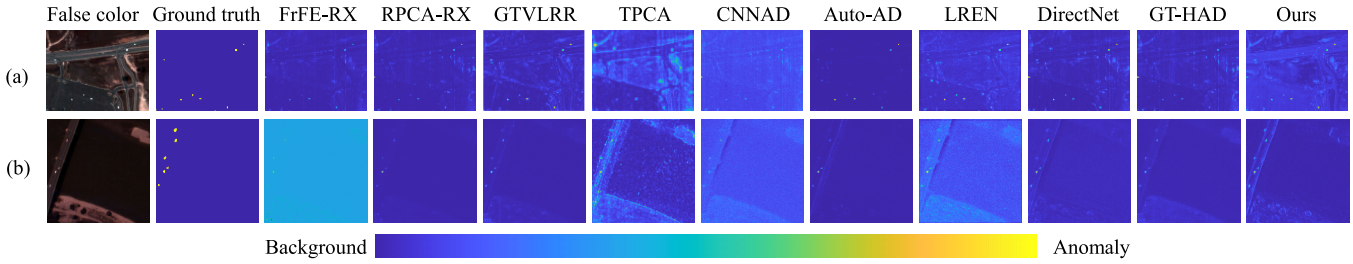


Fig. 12. Anomaly detection results across additional two public datasets using various detectors. (a) HYDICE. (b) Pavia.

Fig. 10(b), several methods such as FrFE-RX, RPCA-RX, GTVLRR, Auto-AD, and DirectNet show limited distinction between the background and anomaly regions. Although TPCA, CNNAD, LREN, and GT-HAD manage to achieve some degree of separation between these regions, the background boxes remain relatively wide, indicating that background suppression is not fully effective. In contrast, SSHAD demonstrates superior performance by maximizing the separation between the background and anomaly regions while minimizing the width of the background box.

D. Ablation Experiments

This section presents comprehensive ablation experiments to assess the various components in SSHAD. After removing all MFFMs as illustrated in Fig. 1(a), we retain the encoder and decoder modules, referred to as AE. Furthermore, we integrate the SSSM and FDSM into the AE, respectively. To ensure a fair comparison, identical settings for the learning rate, batch size, and epochs are applied across all experiments. The comparisons of AUC(P_D, P_F) on six datasets under different combinations are presented in Table V. Clearly, the base architecture AE has limitations in detecting anomalies. However, after integrating the SSSM and FDSM, the anomaly detection accuracy demonstrates significant improvement across all six datasets. The visual ablation comparisons on HSI-I and HSI-II are presented in Fig. 11. We mark the areas in the background that are prone to misidentification as anomalies in red boxes. Notably, AE faces challenges in detecting anomalies, such as detecting some anomalous targets with poor clarity and insufficient background suppression. The AE + SSSM suppresses background elements by capturing a global receptive field and learning sparse spatial characteristics. Furthermore, AE + FDSM enhances the highlighting of anomalous areas by focusing on inter-spectral similarities, thereby improving local feature aggregation. When both SSSM and FDSM are combined, the SSHAD model not only suppresses background more effectively but also achieves sharper detection of anomalies across varying spatial and spectral characteristics. This clearly demonstrates the complementary roles of the SSSM and FDSM modules in enhancing overall anomaly detection performance.

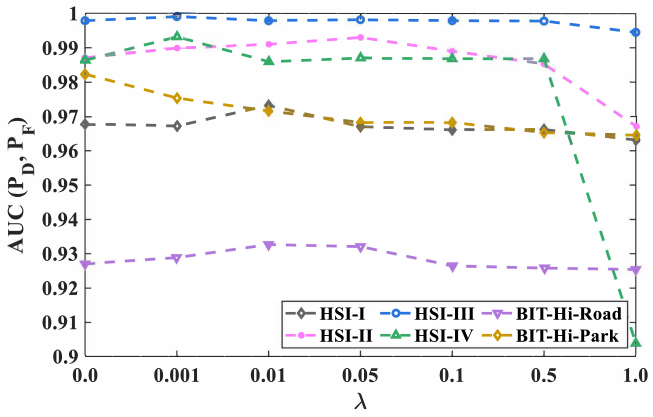


Fig. 13. Effect of balance parameter λ on AUC(P_D, P_F).

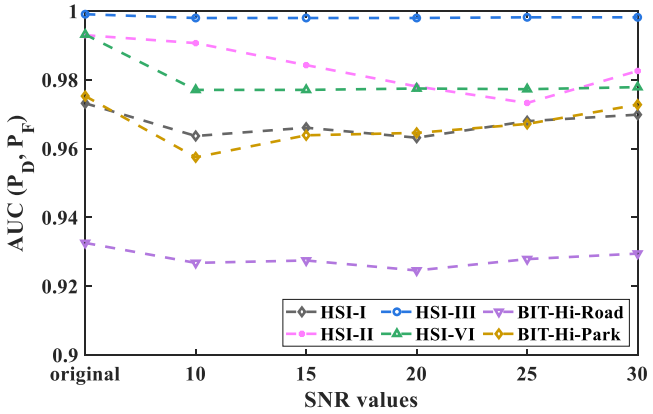


Fig. 14. Robustness performance of SSHAD on the six datasets.

The performance of SSHAD is influenced by the configurations of the MFFMs and SS-blocks. As shown in Table VI,

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TABLE IX
COMPUTATION TIMES COMPARISON FOR DIFFERENT DETECTORS (UNIT: SECONDS). BOLDFACE INDICATES THE SHORTEST TIME, AND UNDERLINING INDICATES THE SECOND-SHORTEST TIME

Datasets	FrFE-RX	RPCA-RX	GTVLRR	TPCA	CNNAD	Auto-AD	LREN	DirectNet	GT-HAD	Ours
HSI-I	17.79	**9.20**	203.91	19.37	46.73	33.59	67.75	827.54	27.65	30.53
HSI-II	22.32	**10.81**	290.60	18.48	44.88	32.78	48.88	757.54	27.43	29.70
HSI-III	16.62	**11.27**	178.53	17.89	45.59	29.39	44.59	849.73	27.53	24.80
HSI-IV	16.88	**9.28**	141.83	18.88	45.68	34.80	47.49	811.23	26.04	29.67
UHAD-U-I	25.64	**8.17**	332.53	31.34	60.80	45.78	70.95	2563.75	166.56	35.95
UHAD-U-II	23.39	**10.71**	305.72	33.75	57.38	53.80	77.43	2675.94	162.61	37.53
Average	20.44	**9.90**	242.18	23.28	50.17	38.35	59.15	1414.28	87.56	31.36

TABLE X
MODEL COMPLEXITY COMPARISON FOR DEEP LEARNING-BASED DETECTORS ON HSI-III

Model	CNNAD	Auto-AD	LREN	DirectNet	GT-HAD	Ours
Params/M	3.56	3.25	0.86	0.49	0.26	0.224
FLOPs/G	89.03	5.98	0.19	49.6	2.64	0.218
Time/s	45.59	29.39	44.59	849.73	27.53	24.80

we conduct a comparative analysis of various combinations of MFFMs and SS-blocks on the HSI-III dataset. Notably, when both p and m are set to 3, the model achieves an AUC(P_D, P_F) of 0.9970, highlighting its effectiveness in anomaly detection. This configuration demonstrates a low parameters (Params) count and efficient floating-point operations (FLOPs). Thus, the selection of both p and m as three balances detection performance and computational efficiency, making it a suitable choice. To verify the effectiveness of Conv 1×1 in the key module SS-block, we conduct comparative experiments on the HSI-III dataset. The results corresponding to different convolution scales and their respective padding are illustrated in Table VII. As the kernel size increases, AUC(P_D, P_F) exhibits a relatively decreasing trend, while Params and FLOPs show an increasing trend. These findings highlight the efficiency of 1×1 convolutions in balancing high detection performance with low computational complexity.

E. Component Analysis

1) Generalization Analysis of Different Sensors: To assess the generalization performance of the proposed method across different sensors, experiments are carried out on another two public HSI datasets, shown in Fig. 12 and Table VIII. The HYDICE dataset, collected by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) airborne sensor, captures an urban area in California, USA. It spans the spectral range from 400 to 2500 nm, with dimensions of 80×100 pixels and a spatial resolution of 2 m/pixel. This scene includes multiple land cover types, with 17 pixels identified as anomalies, such as vehicles, roofs, and other distinct objects. The Pavia dataset was acquired using the Reflective Optics System Imaging Spectrometer (ROSIS) sensor and depicts part of Pavia, Italy. It includes dimensions of 150×150 pixels with a spatial resolution of 1.3 m/pixel, covering the spectral range from 430 to 860 nm with 102 bands after excluding unsuitable bands. Key anomalies in this dataset include vehicles and small objects distributed across urban structures, such as bridges and water regions. From the detection maps in Fig. 12 and AUC(P_D, P_F) comparisons in Table VIII, the proposed SSHAD method demonstrates robust performance on datasets from both HYDICE and ROSIS sensors, indicating satisfactory detection capabilities. Together with the results from datasets obtained with the AVIRIS and HyperspecI sensors aforementioned, these findings support the generalization capability of the proposed method across diverse sensors.

2) Analysis of Balance Parameter λ: This section evaluates the impact of the parameter λ, which balances adaptive mse loss and FFL. Comparative experiments are conducted on six datasets, using λ values from the set {0.0, 0.001, 0.01, 0.005, 0.1, 0.5, 1.0}. Fig. 13 illustrates how different λ values affect detection accuracy across the datasets. Results show that increasing λ generally improves detection accuracy, emphasizing the role of FFL in accurately reconstructing the background. However, when λ is set too high, detection accuracy significantly decreases as it becomes difficult to distinguish between the background and anomalies.

3) Analysis of Running Time: Furthermore, we assess the computational times of various detectors across the four datasets, detailed in Table IX. RPCA-RX exhibits the shortest computation time, attributed to its direct calculation of the background variance matrix. In contrast, GTVLRR requires the longest computation time due to its intricate processes involving rank estimation and coefficient matrix regularization. Importantly, SSHAD strikes an optimal balance between detection precision and computational efficiency, demonstrating highly promising results.

4) Analysis of Model Complexity: Table X presents a comparison of Params, FLOPs, and runtime for various deep learning-based methods evaluated on the HSI-III dataset. Notably, SSHAD significantly reduces both Params and FLOPs compared to CNNAD and Auto-AD. Although

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LREN has slightly lower FLOPs than SSHAD, it exhibits considerably higher Params and runtime due to its optimization process. DirectNet is efficient in terms of Params but suffers from high FLOPs and runtime. In contrast, GT-HAD provides a balanced performance across Params, FLOPs, and runtime. However, SSHAD offers a more favorable trade-off between model complexity and execution time. Overall, SSHAD achieves satisfactory detection accuracy while effectively reducing model complexity, demonstrating its potential for creating lightweight models.

5) Analysis of Robustness: To investigate the SSHAD method's ability to withstand noise, experiments are performed across six datasets by introducing varying levels of zero-mean Gaussian noise (SNR of 0, 10, 15, 20, 25, and 30 dB). The results, depicted in Fig. 14, show the AUC(P_D, P_F) values of SSHAD under different noise conditions. The findings indicate that as the SNR increases, detection accuracy improves correspondingly. When the SNR exceeds 15 dB, the AUC(P_D, P_F) values stabilize, highlighting the SSHAD method's robustness to noise.

F. Discussion

The proposed SSHAD method is designed for HAD tasks in various fields, including airborne remote sensing, urban monitoring, environmental monitoring, and precision agriculture. These scenarios commonly involve complex background structures and sparse anomalies. In these applications, anomalies are typically objects or materials with distinct spectral signatures that differ sharply from the surrounding environment, which allows them to be distinguished through residual errors. While SSHAD generally demonstrates robust performance, the background suppression needs promotion. This limitation may stem from the frequency attention mechanism in the FDSM module. While it enhances high-frequency anomaly features, strong low-frequency background signals may result in incomplete noise suppression during reconstruction. A possible solution is to optimize filtering mechanisms to more selectively suppress nonanomalous high-frequency signals.

IV. Conclusion

In this study, we present a novel SSHAD for HAD, which effectively integrates spatial sparsity and interspectral similarity through an adaptive dual-domain learning framework. The proposed SSHAD explores both spectral and spatial information while performing frequency-division processing of the background and anomalies. By incorporating the SSSM and the FDSM in parallel within an SS-block, we improve the model's capacity to detect anomalies while maintaining linear computational complexity. Experiments conducted on four public datasets and two UAV-borne HSI datasets validate the effectiveness and generalizability of SSHAD across diverse real-world scenarios. Results indicate that SSHAD not only outperforms several state-of-the-art methods but also achieves lower computational costs. Future work will focus on exploring adaptive region selection techniques and investigating advanced uncertainty modeling approaches to enhance the model's efficiency and interpretability.

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